

Understanding Sampling Frequency (fs)

In signal processing, **fs** controls how frequently a continuous-time signal is sampled. It determines how many samples are taken every second.

What is fs?

```
fs = 1000;
```

This means **sampling frequency = 1000 samples per second**. The unit is Hz (samples/second).

Examples:

- $fs = 1000 \rightarrow 1000$ samples every second
- $fs = 10 \rightarrow 10$ samples every second
- $fs = 100 \rightarrow 100$ samples every second

Sampling period

Your time vector is:

```
t = 0:1/fs:1-1/fs;
```

The quantity $1/fs$ is the time interval between consecutive samples, usually called the sampling period T_s :

$$T_s = 1 / fs$$

fs	Sampling period (1/fs)	Samples in 1 second
1000 Hz	0.001 s = 1 ms	1000
100 Hz	0.01 s = 10 ms	100
10 Hz	0.1 s = 100 ms	10

Your 2-Hz square wave

```
f = 2;
```

The signal frequency is 2 Hz, so its period is:

$$T = 1/f = 1/2 = 0.5 \text{ s}$$

With fs = 1000 Hz

The number of samples per cycle is:

$$1000 / 2 = 500 \text{ samples per cycle}$$

This is a very dense sampling, so the plotted signal has a fine numerical representation.

With fs = 10 Hz

The number of samples per cycle is:

$$10 / 2 = 5 \text{ samples per cycle}$$

Now each cycle is represented by only five samples, so the numerical representation is much coarser.

Nyquist sampling criterion

A fundamental idea in digital signal processing is the Nyquist criterion. For a signal whose highest frequency component is $f_{\max}$, the sampling frequency should generally satisfy:

$$f_s > 2 f_{\max}$$

For your 2-Hz signal:

$$f_s > 2(2) = 4 \text{ Hz}$$

Therefore, $f_s = 10 \text{ Hz}$ is technically above the basic Nyquist limit for a 2-Hz signal, while $f_s = 1000 \text{ Hz}$ provides many more samples and therefore a much finer numerical representation.

Key takeaway

Higher f_s → more samples per second → finer representation.

Lower f_s → fewer samples per second → coarser representation.

This concept is fundamental to DFT, FFT, sampling, and digital signal processing.